VALAY BUNDELE

Research Interests

Vision-Language Models, Video Object Tracking, Self-supervised Learning, Efficient Vision Transformers, Digital Pathology, Surgical Video Analysis

Education

International Max Planck Research School for Intelligent Systems — Tuebingen, Germany
PhD student in the Computer Graphics and Vision lab at University of Tuebingen — *Sept 2023 – Present*

- Thesis (tentative): Robust Tracking Through Occlusions
- Advisor: Prof. Hendrik Lensch

Indian Institute of Technology Bombay — Mumbai, India
B.Tech in Energy Engineering with M.Tech in AI and Data Science — *July 2018 – July 2023*

- Cumulative GPA: **8.77/10.0**
- Minor Degree in Computer Science and Engineering

Aklank Public School(CBSE) — Kota, India
Intermediate/+2 — *Percentage: 91.40%*

Publications

** denotes equal contribution*

1. **V. Bundele**, S. Agrawal, M. Hosseinzadeh, N. Nguyen, H. Lensch; 'When Trackers Fail: VLM-Guided Verification and Recovery for Robust Video Object Segmentation', Advances in Neural Information Processing Systems (NeurIPS) 2026

2. A. Stammer, **V. Bundele**, M. Hosseinzadeh, H. Lensch; 'Learning Where to Focus: Self-Supervised Multi-Scale ViTs for Histopathology', DAGM German Conference on Pattern Recognition (GCPR) 2026

3. **V. Bundele**, M. Hosseinzadeh, H. Lensch; 'ReMeDI: Refined Memory for Disambiguation of Identities with SAM3 in Surgical Segmentation', International Conference on Medical Image Computing And Computer Assisted Intervention (MICCAI) 2026

4. **V. Bundele***, K. Sarıtaş*, B. Kargi*, O. Çal*, K. Tezören*, Z. Ghaderi, H. Lensch; 'Evaluating Self-Supervised Learning in Medical Imaging: A Benchmark for Robustness, Generalizability, and Multi-Domain Impact', British Machine Vision Conference (BMVC) 2025

5. E. Grosskopf*, **V. Bundele***, M. Hosseinzadeh*, H. Lensch; 'HistDiST: Histopathological Diffusion-based Stain Transfer', DAGM German Conference on Pattern Recognition (GCPR) 2025

6. **V. Bundele**, M. Hosseinzadeh, H. Lensch; 'You Are Your Best Teacher: Semi-Supervised Surgical Point Tracking with Cycle-Consistent Self-Distillation', SynData4CV Workshop at Computer Vision and Pattern Recognition (CVPR) 2025

7. P. Bele, **V. Bundele**, A. Bhattacharya, A. Jha, G. Roig, B. Banerjee; 'Learning Class and Domain Augmentations for Single-Source Open-Domain Generalization', IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024

8. **V. Bundele**, M. Bhupati, B. Banerjee, A. Grover; 'Scaling Vision-and-Language Navigation with Sub-Optimal Trajectories', Transactions of Machine Learning Research (TMLR) 2024

9. S. Nasser, S. Shamsi, **V. Bundele**, B. Garg, A. Sethi; 'Perceptual cGAN for MRI Super-resolution', IEEE Engineering in Medicine and Biology Conference (EMBC) 2022

10. D. Pal, **V. Bundele**, R. Sharma, B. Banerjee, Y. Jeppu; 'Few-Shot Open-Set Recognition of Hyperspectral Images with Outlier Calibration Network', IEEE Winter Conference on Applications of Computer Vision (WACV) 2022

11. D. Pal*, **V. Bundele***, B. Banerjee, Y. Jeppu; 'SPN: Stable Prototypical Network for Few-Shot Learning-Based Hyperspectral Image Classification', IEEE Geoscience and Remote Sensing Letters, 19, 1-5.

Professional and Teaching Experience

Software Development Intern | Amazon Web Services *May – July 2021*

Interned in the Aurora CP team which works on AWS Relational Database Management System

- Worked on graph-based **integrated testing** framework of Relational Database Service to detect destructive paths
- Utilized **Java-callgraph** to generate a static call graph and thereby get the function calls made in each function
- Implemented a **depth-first search** based approach to recursively check for annotation on each function using **Reflection API** and classified paths having function calls with a specific annotation as destructive

Teaching Assistantships

Facilitating smooth course organization, grading papers, mentoring students, conducting tutorials and help sessions

- **TA:** Computer Graphics, Prof. Hendrik Lensch *Oct - Mar 2025*
- **TA:** Practical Course for Vision and NLP, Prof. Hendrik Lensch *Apr - July 2024*
- **TA:** GNR 638 - Machine Learning for Remote Sensing II, Prof. Biplab Banerjee *Aug - Nov 2021*
- **Head TA:** GNR 638 - Machine Learning for Remote Sensing II, Prof. Biplab Banerjee *Aug - Nov 2022*

Academic Reviewing

Peer-reviewer for top-tier conferences and journals in computer vision and machine learning

- Reviewer for: NeurIPS, CVPR, ECCV, WACV, BMVC, MICCAI, TMLR, Pattern Recognition

Technical Skills

Programming Languages: C++, Python
Libraries/Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Numpy, Pandas
Software: Microsoft Office, LaTeX, Git, VS Code

Relevant Coursework

Machine Learning: Introduction to Machine Learning, Advanced ML, Medical Image Computing, Foundations of Intelligent and Learning Agents, Learning with Graphs, Deep Learning : Theory and Practice

Computer Science: Operating Systems, Logic in CS, Data Structures & Algorithms, Design & Analysis of Algorithms

Mathematics: Calculus, Linear Algebra, Differential Equations, Numerical Analysis

Select Mentoring and Leadership Roles

Software Subsystem Head | Team Rakshak *Apr 2021 - July 2022*

Head of the software subsystem of Team Rakshak which works on design and development of UAVs

- Led and managed a team of **10 members** to develop algorithms for the tasks of Object Detection, Object Recognition, and Mapping of surveyed areas for **AUVSI SUAS** competition held in **Maryland, USA**
- Employed YOLOv5 for detection and trained classifiers having **ResNet** backbone to identify object characteristics
- Implemented **PCA** to find the orientation of objects and performed image stitching using **SIFT** to generate a map

Coordinator at Mood Indigo, IIT Bombay | Competitions and LYP *Aug - Dec 2019*

Asia's largest college cultural festival; 146,000+ foot-fall with 240+ events

- Member of the team that ideated and revamped the governing rules of **5+** competitions in various genres
- Assisted in conceptualizing and organizing Multicity Competitions in **5+** cities, thus increased outreach
- Invited renowned artists across various cultural genres as judges for the competitions on a **strict zero budget**

Academic Achievements and Extracurricular Activities

Achievements	• Secured **All India Rank 2162** in Joint Entrance Examination among **1.1 million** candidates (2018) • Among **Top 1%** students (nationally) in first stage of IJSO among 46,688 candidates (2014-15) • Secured **All India Rank 18** in STaRT examination conducted by Resonance Eduventures (2015) • Obtained International Rank 193 in National Science Olympiad conducted by SOF (2013)
Mentorship	• Mentored 2 students at Summer Of Science 2020 to understand concepts of Deep Learning • Guided 8 students on the project titled "Image Colorization" - IITB Seasons of Code 2021 • Mentored 5 students on the project titled "Image Editing" - IITB Seasons of Code 2022
Others	• Assisted in successful execution of the speaker sessions at the **E-Summit** conducted by E-Cell (2020) • Completed a biathlon that consisted of 5 kms running and 7 kms cycling (2019) • Successfully completed a year-long guitar training under National Sports Organization (2018-19) • Participated in Dainik Bhaskar's National Level Newspaper Making Competition (2014)